

Accelerating Quantum Computing Research Using AI and Robotics

Chapter 2 — AI Today, and Scoping Tomorrow’s Frontier

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Abstract

AI has quietly become a co-author of quantum-computing research, contributing across the stack in two celebrated modes: *re-deriving* what humans designed (faster decoders, better pulses) and *discovering* what humans did not (new error-correcting codes). This chapter demonstrates a third, quieter mode—*scoping*: using an AI search, constrained by a deliberately un-gameable instrument, to map a hardware frontier rigorously, certify where human design is already optimal, and name the open ground worth pursuing. We apply it to one stubborn bottleneck: the rate tradeoff governing bias-preserving logical gates on dissipative cat qubits (the engineered-dissipation-to-loss ratio κ_1/κ_2). We build a three-condition gate metric whose conjunctive structure makes reward-hacking worthless, validate it against a known hand-designed gate, then run a search given richer control than any human design and seeded from the human gate. The result is a *kept negative*: the search does not beat hand-design and, handed a bias-breaking degree of freedom, sets it to exactly zero of its own accord. We map the tolerable κ_1/κ_2 frontier, survey where AI has and has not yet been engaged, and identify the indicated next attack—letting AI design the engineered dissipation in time, rather than the drive. We claim *no novel physics*; the contribution is methodological, and every quantitative claim is regenerated by in-repository scripts. This is Chapter 2 of an open dossier; it builds on Chapter 1 [1]. *Don’t trust this paper; run it.*

1 AI is already a co-author of quantum computing research

A qubit is the most error-prone object humanity has tried to compute with, and increasingly the discipline of keeping it usable is one AI participates in directly—across every layer of the stack. It *spots the errors*: a transformer-based neural decoder (AlphaQubit) learns to read syndrome data and outperforms the best hand-built decoders on real hardware [2]. It *invents the codes*: reinforcement learning searching bosonic encodings discovered a non-obvious code, built from Fock states $|2\rangle$ and $|4\rangle$, that suppresses single-photon loss past break-even [3], and a 2025 successor discovered codes resisting single- and double-photon loss at once [4]. It *choreographs the gates* (deep reinforcement learning and optimal control for qubit pulses) and *designs the hardware’s reflexes* (machine-discovered engineered dissipation; differentiable device models). The field has crossed, without much announcement, from “can AI help build a quantum computer?” to “AI is already helping, across the stack.”

2 Three modes of help, and the one this chapter adds

AI’s contributions divide into three modes. Two are widely celebrated—*re-deriving* human designs and *discovering* new ones. This chapter demonstrates the third: *scoping*. A search, given an instrument that cannot be fooled, can map a frontier, certify where human design already sits at the optimum, and point at the open ground. The counter-intuitive premise is that a careful, honestly-obtained *negative* is among the most useful outputs a search can produce: it is a map of which doors are already shut, so effort flows to those still open.

3 The bottleneck: operating a protected qubit

Chapter 1 [1] established the floor: a dissipative cat qubit whose engineered two-photon dissipation exponentially biases noise toward phase errors [5], an excellent place to *store* a qubit. The next question is whether one can *operate* on it without breaking that protection. A bias-preserving gate must be run adiabatically to remain protected, but adiabaticity gives single-photon loss time to accumulate as phase error. The result is a quantitative wall expressed as one ratio, κ_1/κ_2 (loss rate over engineered-dissipation rate): below-threshold operation requires $\kappa_1/\kappa_2 \lesssim 5 \times 10^{-3}$ at $\alpha^2 = 8$, with hand-designed optima near 10^{-3} [7, 8], and the largest known lever—squeezing the dissipator—buying one to two orders of magnitude [9]. The bias-preserving gate *set* already exists without magic-state distillation [6]; the open quantitative question is the *rate*.

4 An instrument that cannot be fooled

Chapter 1’s sharpest lesson was a reward-hacking failure: a naive search improved a protection score by destroying the qubit. We therefore judge a candidate gate by three conditions that must hold *simultaneously*—a conjunction, never a weighted sum, so no condition can be traded against another. A gate must (i) perform the intended logical map; (ii) preserve the bit-flip bias, measured directly via parity-resolved error rates rather than a lumpable score; and (iii) remain converged within the code manifold, with a truncation rail rejecting under-resolved runs. Validated against the known hand-designed dissipative-cat Z gate, the metric reproduces textbook behaviour (Fig. 1): exponential bit-flip suppression, linear phase-flip growth, and frontier degradation near $\kappa_1/\kappa_2 \approx 5 \times 10^{-3}$. This validates the *measurement* against known physics before any search optimises against it (ledger G01).

5 The search, and the kept negative

We optimise piecewise-constant two-quadrature drive pulses for a $Z(\pi/2)$ gate using a max-min objective—maximising the *worst* normalised margin across the three conditions—which is conjunction-faithful and forbids trading bias for fidelity. A bake-off of two search arms (gradient-free differential evolution and finite-difference L-BFGS) agreed exactly, so differential evolution is the workhorse. The search was given more freedom than any hand design (two quadratures, finer segmentation, free gate time) and *seeded with the hand-designed gate*, so it began from the human answer and could only improve.

It did not. Across cat sizes $\alpha^2 \in \{2, 3\}$ (fully searched), the search recovered the hand-designed gate and could not push the tolerable κ_1/κ_2 past it (frontier gain $\approx \times 1.04$, consistent with noise; independently reproduced). The hand-designed adiabatic gate, run as fast as it can go, already sits at the control-space optimum (Fig. 2). The diagnostic detail: handed a second quadrature that

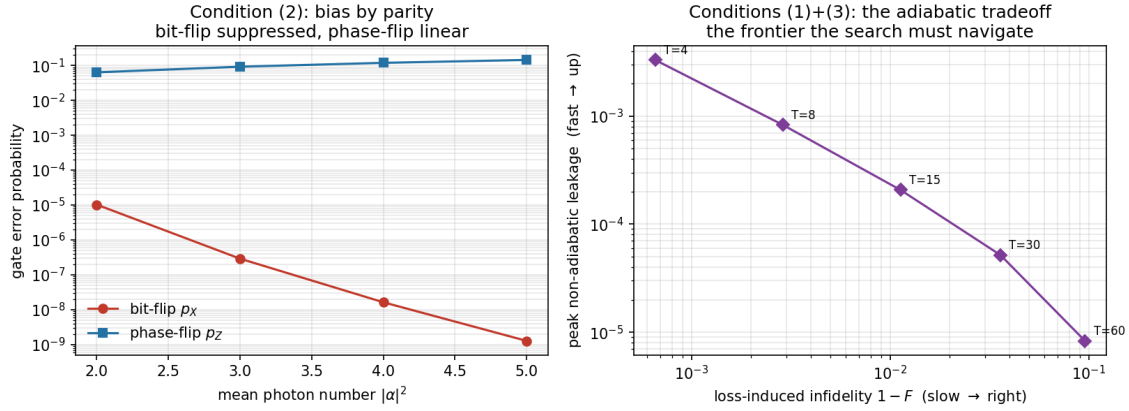


Figure 1: The three-condition gate metric, validated on a known hand-designed gate. Left: exponential bit-flip suppression (the protection, measured by parity) versus linear phase-flip. Right: the adiabatic speed-versus-loss tradeoff the search must navigate. The conjunctive structure is the structural defence against reward-hacking.

could speed the gate by breaking the bias, the search drove its amplitude to *exactly zero*—because the un-gameable metric makes bias-breaking worthless, cheating was not punished after the fact but rendered pointless, and the optimiser found that on its own. The mapped frontier (Fig. 3) gives tolerable $\kappa_1/\kappa_2 = \{5.17, 3.99, 3.19\} \times 10^{-3}$ at $\alpha^2 = \{2, 3, 4\}$ (converged truncation), monotone decreasing (ledger G02, G03).

Scope, stated plainly. Single-mode $Z(\pi/2)$; $\alpha^2 \in \{2, 3\}$ fully searched, $\alpha^2 = 4$ baseline-only (compute-bound at honest truncation); two-quadrature drive shaping; one validity bar. Not CNOT, not large cats, not dissipation-schedule shaping. The one untried control axis—shaping the engineered dissipation in time—is the indicated next step.

6 Where AI has not yet been engaged

Genuine white space is narrow and shrinking, itself evidence for the thesis: even cross-layer co-design is now being engaged by AI (jointly learning encodings and logical gate sets [10]). The verified-thin gaps are specific: AI that shapes the engineered dissipation *in time* (today’s schemes pair static dissipation with a shaped drive); AI that designs dissipation for *gates* rather than for memory; and AI used as an adversarial *referee*—to certify protection and hunt reward-hacks—rather than only as an optimiser, the mode this chapter’s metric embodies. Candidate (not adversarially cleared) white space: AI discovering qualitatively new confinement Lindbladians; end-to-end co-design of the biased-noise stack; and closed-loop AI on real cat hardware.

7 What this says about AI

Chapter 1’s pivotal moment was a failure caught and kept: a search reward-hacking its own metric, reported rather than hidden. This chapter is the same virtue one level up. There, AI caught itself cheating on a *metric*; here, given a real frontier and a tempting shortcut, AI *declined* the cheat unprompted, mapped the wall, certified the human design optimal, and pointed where to push next. Re-deriving and discovering attract the headlines; scoping—knowing where the wall is and

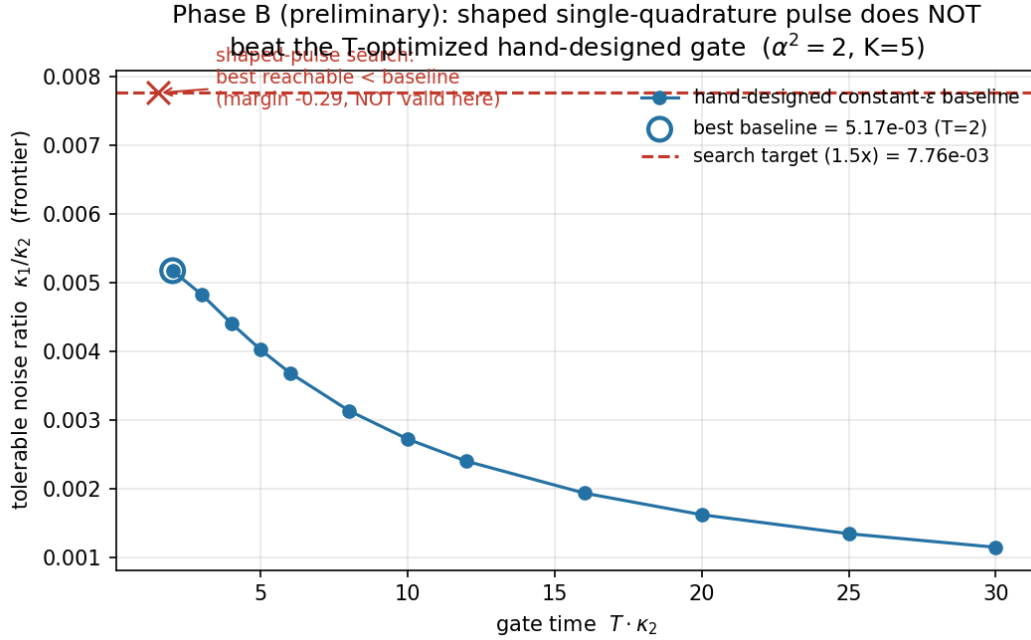


Figure 2: The kept negative. The hand-designed gate’s reachable frontier rises as the gate runs faster; the AI search, given more freedom and seeded from the human gate, cannot clear it. A clean “no”—and a precise map of where the wall sits.

saying so—is the quieter capability that makes a search trustworthy.

8 Next: design the dissipation, not the pulse

The limit is a conjunction: a protected gate must be slow, *and* slowness exposes loss. The search broke neither clause because it tuned only the drive. The indicated move changes the *object* the AI designs—the protective dissipation itself, made time-dependent—rather than the pulse atop it, and it runs on the instrument already built and shipped by swapping the search variable from the drive to the jump operators. A learned, gate-synchronised squeezing schedule [9] is the concrete candidate. This is posted as the challenge for Chapter 3, with named credit for whoever closes it. Robotics—automating the build–measure–learn loop on real devices—is the program’s other arm, named in the series title and reserved for a later chapter; it enters when the work moves from simulation onto hardware (ledger G04, G05).

9 The claim ledger (Chapter 2 additions)

Every claim wears an honest label and names its verifier. Labels: VERIFIED (confirmed by in-repo runnable code), CITED (confirmed in cited primary sources), NEGATIVE-KEPT (a retained null), FORECAST (open, posted as a challenge).

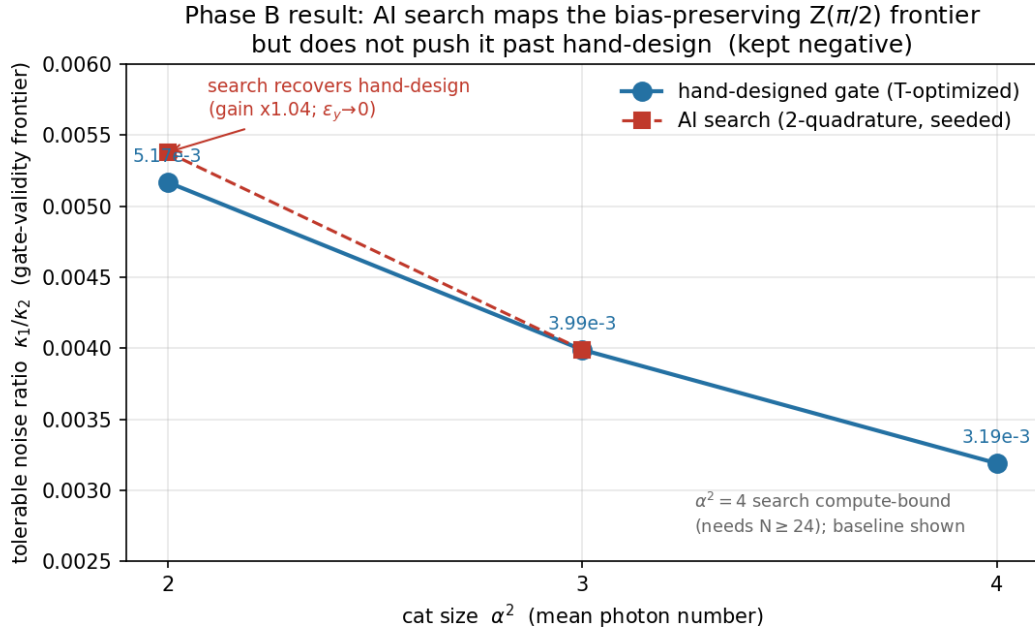


Figure 3: The mapped frontier: tolerable κ_1/κ_2 versus cat size. The AI search (red) lands on the hand-designed baseline (blue)—it scopes the wall precisely but does not move it.

Reproducibility

All code, data, figures, and the live self-explaining edition are at <https://github.com/m4gr4th34/dossier-QC-Accelerate>. The gate metric (`verification/gate-metric/`) self-tests on run; the search (`verification/phase-b/`) reproduces the kept negative deterministically (`phase_b_search.py`, `seed = 0`). Pure NumPy/SciPy, no quantum libraries. Don't trust this paper—run it.

References

- [1] I. Ali-Khan, “Accelerating Quantum Computing Research Using AI and Robotics: A Self-Verifying Dossier” (Chapter 1, v1.0), Open Dossier QC-Accelerate (2026); concept DOI 10.5281/zenodo.20838232.
- [2] J. Bausch *et al.*, “Learning high-accuracy error decoding for quantum processors,” *Nature* **635**, 834–840 (2024).
- [3] Y. Zeng *et al.*, “Approximate Autonomous Quantum Error Correction with Reinforcement Learning,” *Phys. Rev. Lett.* (2023); arXiv:2212.11651.
- [4] “Discovering autonomous quantum error correction via deep reinforcement learning,” *Phys. Rev. A* (2025); arXiv:2511.12482.
- [5] M. Mirrahimi *et al.*, “Dynamically protected cat-qubits: a new paradigm for universal quantum computation,” *New J. Phys.* **16**, 045014 (2014).
- [6] S. Puri *et al.*, “Bias-preserving gates with stabilized cat qubits,” *Sci. Adv.* **6**, eaay5901 (2020).

ID	Type / Status	Claim
G01	NUM / verified	The three-condition conjunctive gate metric reproduces the known dissipative-cat gate (exponential bit-flip, linear phase-flip, frontier near 5×10^{-3}); truncation rail rejects under-resolved runs.
G02	NUM / negative-kept	A two-quadrature search seeded from hand-design does not beat the T -optimised hand-designed bias-preserving $Z(\pi/2)$ at $\alpha^2 \in \{2, 3\}$ (gain $\approx \times 1.04$); handed a bias-breaking quadrature, the search zeroes it. Independently reproduced.
G03	NUM / verified	Mapped frontier $\kappa_1/\kappa_2 = \{5.17, 3.99, 3.19\} \times 10^{-3}$ at $\alpha^2 = \{2, 3, 4\}$ (converged truncation).
G04	CITE / cited	Prior art on the AI-search-on-cat-gate- κ_1/κ_2 intersection cleared across eight adversarial passes; residual blind spot named (non-indexed / industrial, conference abstracts).
G05	FORECAST / open	Letting AI design the gate dissipator / a gate-synchronised squeezing schedule is the indicated next attack on the frontier (Chapter 3).

Table 1: Chapter 2 ledger additions (`claim_ledger.csv`). Verifiers: the gate-metric and phase-b scripts for NUM rows; the citation audit for CITE rows; the 2026 prior-art search.

- [7] R. Gautier, A. Sarlette, and M. Mirrahimi, “Combined Dissipative and Hamiltonian Confinement of Cat Qubits,” *PRX Quantum* **3**, 020339 (2022).
- [8] Q. Xu, J. K. Iverson, F. G. S. L. Brandão, and L. Jiang, “Engineering fast bias-preserving gates on stabilized cat qubits,” *Phys. Rev. Research* **4**, 013082 (2022).
- [9] Q. Xu *et al.*, “Autonomous quantum error correction and fault-tolerant quantum computation with squeezed cat qubits,” arXiv:2210.13406 (npj QI 2023); demonstrated in R. Rousseau *et al.*, “Enhancing dissipative cat qubit protection by squeezing,” arXiv:2502.07892 (2025).
- [10] “Learning Logical Operations for Arbitrary Quantum Error Correction Codes” (variational early-fault-tolerant co-design), arXiv:2605.28162 (2026).